

L5 : Interpretations of Ordinary Least Squares (OLS) regression

Today's agenda:

- Recap of OLS
- Geometric interpretation
- Statistical interpretation

Recall that linear regression is a supervised learning approach that tries to fit a linear input-output relationship.

$$\hat{y} = f(x) = \hat{\beta}_0 + \left\langle \begin{pmatrix} \hat{\beta}_1 \\ \vdots \\ \hat{\beta}_p \end{pmatrix}, x \right\rangle$$

$$= \left\langle \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \vdots \\ \hat{\beta}_p \end{pmatrix}, \begin{pmatrix} 1 \\ x \end{pmatrix} \right\rangle$$

We fit $\hat{\beta}$ by minimizing residual sum of squares (RSS).

↳ Ordinary least squares (OLS)

$\hat{\beta}$ minimizes $\|Y - X\beta\|^2$

$$\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} \quad \begin{bmatrix} 1 & x_{1T} \\ 1 & x_{2T} \\ \vdots & \vdots \\ 1 & x_{nT} \end{bmatrix} \quad \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{pmatrix}_{(p+1) \times 1}$$

$n \times (p+1)$

★ If columns of X are linearly independent, then

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

$$\Rightarrow \hat{y} = X\hat{\beta} = \underbrace{X(X^T X)^{-1} X^T}_{\text{hat matrix } H} y$$

Aside: While our assumption will be by default that X has full ^{col.} rank, it is also possible ~~to~~ to handle the case where X has linearly dependent columns.

In this case, there are infinitely many solutions for minimizing RSS.

One special solution is:

$$\hat{\beta} = X^+ y$$

pseudo-inverse of X

Geometric interpretation

OLS minimizes $\|y - X\beta\|^2$

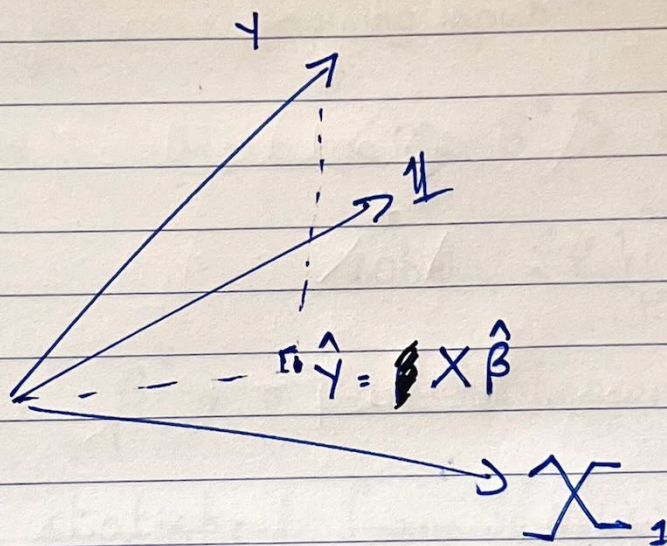
$$\begin{bmatrix} 1 & X_1 & X_2 & \dots & X_p \end{bmatrix} \begin{pmatrix} \beta_0 \\ \vdots \\ \beta_p \end{pmatrix}$$

$$\Rightarrow X\beta = \beta_0 \mathbb{1} + \beta_1 X_1 + \dots + \beta_p X_p$$

is a linear combination of the columns of X .

$\Rightarrow$ OLS seeks to find that linear combination of the columns of X that "best approximates" Y

$\Rightarrow$ OLS projects Y on to subspace spanned by columns of X .



★ When X has full col. rank, the hat matrix performs this projection!

$$\hat{Y} = \left(X(X^T X)^{-1} X^T \right) Y$$

$$= H Y$$

is the projection of Y on subspace spanned by columns of X .

Statistical Interpretation

So far, we ~~we~~ have only done empirical risk minimization (ERM), a purely optimization-based approach.

We now impose a statistical model on the data for analysis & interpretation.

Assume $(x_1, x_2, \dots, x_n)$ are fixed (deterministic),
training inputs

$$\text{and } y_i = \langle x_i, \beta^* \rangle + \underbrace{\epsilon_i}_{\text{random noise}},$$

where $\{\epsilon_i\}_{i=1}^n$ are iid $N(0, \sigma^2)$.

Note: β^* & σ^2 are unknown here.

In this statistical framework, we can view $\hat{\beta}$ as our estimate of β^* , given the training data.

The most natural estimator is of-course the MLE (maximum likelihood estimator): it maximizes the likelihood function.

$$\mathcal{L}(\beta) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y_i - \langle x_i, \beta \rangle)^2}{2\sigma^2}}$$

$$\Rightarrow \mathcal{L}(\beta) = \left(\frac{1}{\sqrt{2\pi}\sigma^2} \right)^n \exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \langle x_i, \beta \rangle)^2 \right\}$$

Therefore,

maximizing $\mathcal{L}(\beta)$



minimizing $\sum (y_i - \langle x_i, \beta \rangle)^2$



OLS

Moral: The same OLS solⁿ as before also admits an ~~more~~ interpretation as the MLE in a statistical sense, assuming i) an underlying linear model, ii) Gaussian noise.

The statistical framework also allows us to reason about bias & variance.

Note: $\hat{\beta} = (X^T X)^{-1} X^T Y,$

where $Y = X\beta^* + \begin{pmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{pmatrix}$

$$\begin{aligned}
 \therefore E[\hat{\beta}] &= (X^T X)^{-1} X^T E[Y] \\
 &= (X^T X)^{-1} X^T X \beta^* \\
 &= \beta^*
 \end{aligned}$$

Moral : MLE estimate is unbiased.

Similarly, it can be shown:

$$\text{Cov}(\hat{\beta}^*) = \sigma^2 (X^T X)^{-1}.$$